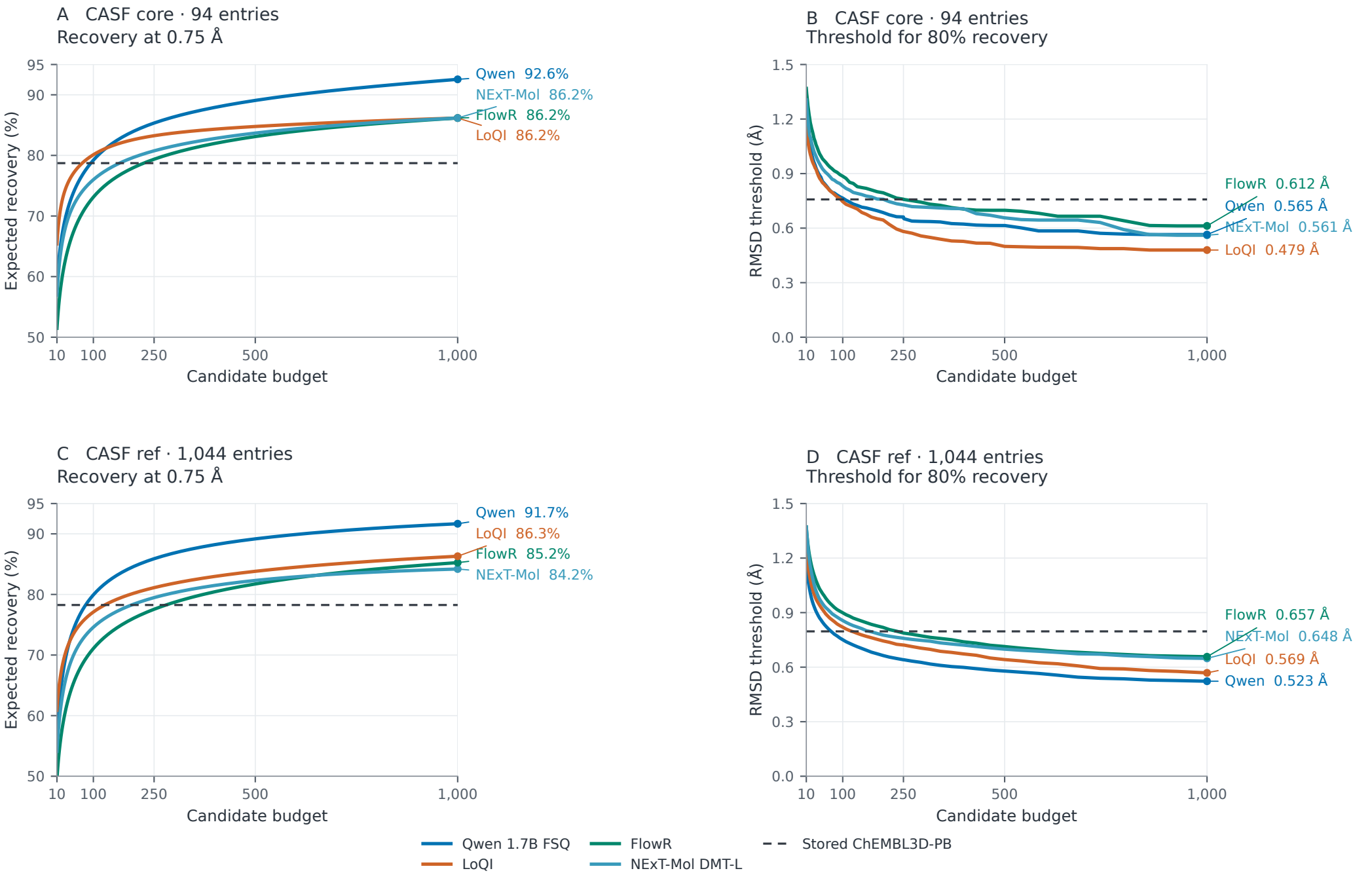Sampling budget, recovery, and matching tolerance